

THE FULL ADJOINT-GENERATED CASE OF GINIBRE'S CONDITION Q_3 PART III: SIGNED-PRODUCT COMPLETION

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ABSTRACT. Let G be a compact connected Lie group with simple Lie algebra and let $\mathcal{Q}_G$ be the closed multiplicative convex cone generated by its real adjoint character. We reduce Ginibre's full condition Q_3 on $\mathcal{Q}_G$ to an explicit two-parameter moment hierarchy and close that hierarchy for every simple type. Parts I–II prove the two-minus row. The present compact guide states the exact higher-minus reduction, the bounded-Littlewood moment contract, and the exhaustive classification certificate. Detailed tail estimates, finite ledgers, verifier semantics, and replay instructions are in the 35-page formal and computational supplement `full_q3_extension.pdf`, included as the final component of the unified submission; that supplement is part of the proof, not an optional derivation archive.

Remark 0.1 (Unified manuscript architecture). The unified manuscript `submission.pdf` contains Parts I–II (`paper.pdf`), this compact Part III guide (`full_q3_main.pdf`), and the complete formal and computational supplement `full_q3_extension.pdf`. The artifact manifest `publication_artifacts.sha256` binds all three components and the unified manuscript. References below to “the supplement” mean that included final component. The optional `paper_full.pdf` development archive is not load-bearing.

1. REDUCTION TO THE HIGHER-MINUS HIERARCHY

Let μ_G be normalized Haar measure, and let $X = \chi_{\text{ad}}(g)$ and $Y = \chi_{\text{ad}}(h)$ for independent Haar variables.

Definition 1.1. Set

$$\mathcal{Q}_G = \left\{ \sum_{j=0}^d c_j \chi_{\text{ad}}^j : d \geq 0, c_j \geq 0 \right\}^{\|\cdot\|_\infty}, \quad I_{a,n}^G = \mathbf{E} (X - Y)^{2a} (X + Y)^n.$$

Every member of $\mathcal{Q}_G$ is a real continuous positive-definite function: characters are positive definite, and products, nonnegative sums, and uniform limits preserve that property.

Proposition 1.2 (Exact reduction and cone promotion). *Ginibre's condition Q_3 holds on $\mathcal{Q}_G$ if and only if $I_{a,n}^G \geq 0$ for all $a, n \geq 0$. Only $a \geq 1$ and odd n are nonautomatic, and Parts I–II prove the complete row $a = 1$.*

Proof. A signed product of the single generator, with r minus factors and n plus factors, is $\mathbf{E}(X - Y)^r (X + Y)^n$. It vanishes for odd r by interchanging X, Y , and equals $I_{a,n}^G$ for $r = 2a$. Even n gives a pointwise nonnegative integrand. For $a = 0$, expansion gives a sum of nonnegative invariant-tensor multiplicities $m_j^G = \dim((\mathfrak{g}_{\mathbb{C}}^{\otimes j})^G)$.

The promotion from χ_{ad} to its generated cone uses

$$(fg)(x) \pm (fg)(y) = \frac{1}{2} \{ (f(x) + f(y))(g(x) \pm g(y)) + (f(x) - f(y))(g(x) \mp g(y)) \}.$$

Iteration, nonnegative linearity, and uniform closure give precisely Ginibre's Proposition 1 [2]. Finally, the $a = 1$ assertion is the main theorem of Parts I–II [1]. $\square$

Lemma 1.3 (Exact moment functional). *For $m_j^G = \mathbf{E} X^j$,*

$$I_{a,n}^G = \sum_{j=0}^{2a} \sum_{k=0}^n (-1)^j \binom{2a}{j} \binom{n}{k} m_{2a-j+k}^G m_{j+n-k}^G.$$

Equivalently, if $M_G(t) = \mathbf{E} e^{tX}$, then

$$\sum_{n \geq 0} I_{a,n}^G \frac{t^n}{n!} = \sum_{j=0}^{2a} (-1)^j \binom{2a}{j} M_G^{(2a-j)}(t) M_G^{(j)}(t).$$

Proof. Expand the two binomials. The generating-function identity follows by expanding only $(X - Y)^{2a}$ after inserting $e^{t(X+Y)}$. $\square$

Proposition 1.4 (Why the cone cannot be enlarged indiscriminately). *For $G = \mathrm{SO}(3)$, the cone of all real continuous positive-definite functions does not satisfy Q_3 . There are four matrix coefficients $f_v(g) = \langle v, gv \rangle$ and the sign pattern $- - ++$ for which the integral is $-8/279$.*

Proof. Each f_v is positive definite since its defining Gram quadratic form is $\|\sum_j z_j g_j v\|^2$. Haar projection on $(\mathbb{R}^3)^{\otimes 2}$ and $(\mathbb{R}^3)^{\otimes 4}$ gives the claimed rational value for the explicit four vectors printed in Supplement §1. Thus this proposition concerns a genuinely larger cone than $\mathcal{Q}_G$. $\square$

2. EXACT FINITE-RANK CONTRACT

The higher-minus proof partitions each simple family into a stable prefix, an exact finite box, and a proved tail. The only substantial finite-rank moment interface not already printed in Parts I–II is the following bounded-Littlewood formula.

Lemma 2.1 (Bounded-Littlewood moment determinants). *In the completed symmetric-function ring, put $e_r = h_r = 0$ for $r < 0$ and*

$$f_r^e = \sum_{k \in \mathbb{Z}} e_k e_{k+r}, \quad f_r^h = \sum_{k \in \mathbb{Z}} h_k h_{k+r}, \quad e = \sum_{k \geq 0} e_k, \quad \bar{e} = \sum_{k \geq 0} (-1)^k e_k.$$

For $b \geq 1$, define

$$\begin{aligned} \mathcal{B}_b &= \frac{1}{2} \{e \det(f_{i-j}^e - f_{i+j-1}^e) + \bar{e} \det(f_{i-j}^e + f_{i+j-1}^e)\}, \\ \mathcal{C}_b &= \det(f_{i-j}^h - f_{i+j}^h), \quad \mathcal{D}_b = \frac{1}{2} \det(f_{i-j}^e + f_{i+j-2}^e), \end{aligned}$$

where all determinants are $b \times b$. Then

$$m_j^{B_b} = [m_{(2j)}] \mathcal{B}_b, \quad m_j^{C_b} = [m_{(2j)}] \mathcal{C}_b, \quad m_j^{D_b} = [m_{(2j)}] \mathcal{D}_b.$$

If F is homogeneous of degree $2j$ and $\pi_2(p_1) = p$, $\pi_2(p_2) = q$, $\pi_2(p_r) = 0$ for $r \geq 3$, then

$$[m_{(2j)}]F = j! \sum_{a=0}^j (2(j-a)-1)!! [p^{2(j-a)} q^a] \pi_2(F).$$

Proof. The orthogonal and symplectic Schur-integral criteria identify the admitted even-column/even-row sectors [4]. The bounded Littlewood identities of Huh–Kim–Krattenthaler–Okada give the three displayed determinants [3]; Hall duality turns their Schur sums into the stated adjoint moments. In the last formula, only $p_1^{2(j-a)} p_2^a$ can contribute to $m_{(2j)}$; counting assignments of the ordered factors gives $j!(2(j-a)-1)!!$. Supplement §8 supplies the degree-by-degree derivation, specializations, and recurrences used by the exact verifier. $\square$

Proposition 2.2 (Exhaustive certificate contract). *For every simple type, every $a \geq 2$, and every odd $n \geq 1$, one has $I_{a,n}^G > 0$. The disjoint stable, finite, and tail regions are those in the following audit summary; the supplement states every endpoint and proves that the regions cover without gaps.*

family	exact finite supplier	stable/tail supplier
A_1	A_1 : 78 values; minimum 16	effective Haar rectangle
A_2	A_2 : 78 values; minimum 72	exact moment cap
A_3, A_4	66 and 136 values; minima 94, 96	exact moment caps
$A_r, r \geq 5$	1,315 nonstable values; minimum 3, 550	stable Schur–Weyl row and uniform tail
G_2, F_4, E_6, E_7, E_8	1,265 values; minimum 36	five exact cap tails
B_b, C_b, D_b	17,862 values; minimum $I_{2,1}^{D_5} = 50$	500 stable identities, exact middle/high and 58 directed low-row tails

The classical finite supplier covers exactly

$$B_2, \dots, B_{21}, \quad C_2, \dots, C_{28}, \quad D_4, \dots, D_{70}.$$

It reconstructs moments through degree 71 by Lemma 2.1, checks all 17,862 values with GMP integers, and uses a 961-bit CRT modulus. An independent checker agrees on the unresolved high-degree moments modulo 21 primes (628 bits, exceeding the 607-bit uniqueness bound) and recomputes the remaining 5,509 inequalities. The least directed low-tail logarithmic margin is 0.0971194051797054477804174... at C_{15} .

Proof. This is the compact statement of Supplement §§2–8. Its type- A proofs combine Schur–Weyl moment formulas, exact cap inequalities, and the stated finite GMP boxes. Its exceptional proof uses exact invariant-tensor moments, five cap tails, and the five exhaustive residual boxes. For the classical families, stable cumulants give the prefix; analytic trace and Weyl cap bounds give the middle and high tails; directed 384-bit interval arithmetic proves all 58 remaining row tails; and Lemma 2.1 supplies the exact finite moments. The supplement prints the row onsets and proves the three regions exhaustive. Each quoted count, minimum, margin, and scope is the output of the named verifier applied to the finite specification stated in the supplement. The mathematical proof consists of that specification, the source-level argument that the verifier implements it, and the exact or outward-rounded execution; the authenticated transcript records the execution and its source identities but is not, by itself, a proof. These three layers yield the proposition. $\square$

3. FULL THEOREM

Theorem 3.1 (Full adjoint-generated Ginibre Q_3). *Let G be any compact connected Lie group with simple Lie algebra. For every finite tuple $f_1, \dots, f_r \in \mathcal{Q}_G$ and signs $\varepsilon_i \in \{+1, -1\}$ containing an even number of minus signs,*

$$\int_{G \times G} \prod_{i=1}^r (f_i(g) + \varepsilon_i f_i(h)) d\mu_G(g) d\mu_G(h) \geq 0.$$

If the number of minus signs is odd, the integral is zero. Thus $\mathcal{Q}_G$ satisfies Ginibre’s condition Q_3 with its full finite-tuple and sign-pattern quantifiers.

Proof. Proposition 1.2 supplies the automatic sectors, imports the two-minus row, and promotes the generator hierarchy to $\mathcal{Q}_G$. Proposition 2.2 supplies every remaining $a \geq 2$, odd- n case in types A, B, C, D and the five exceptional types. Classification and central-cover invariance exhaust all compact connected global forms. $\square$

Remark 3.2 (Reproduction). From the source root, `make publication-artifact-audit` rebuilds and hash-checks the unified manuscript and its three included components; `make full-q3-extension` replays the complete mathematical certificate suite. The CI workflow runs the publication build, the clean room Parts I–II replay, and the Part III replay as separate jobs. Exact commands, trusted-base boundaries, and transcript mappings are in `REPLAY.md` and Supplement §9.

REFERENCES

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